A Hybrid GARCH-Transformer Framework for High-Frequency Value at Risk Estimation in Cryptocurrency Markets: Evidence from Ethereum

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Abstract

The extreme volatility and heavy-tailed distributions inherent in cryptocurrency markets pose significant challenges for traditional risk management frameworks[2]**. To address these limitations in high-frequency trading (HFT) environments, this study introduces an innovative hybrid methodology integrating the GARCH model with a Transformer neural network using Ethereum tick data (ETHUSDC-trades-2026-09-DBlock.csv). By aggregating micro-structure data into volume-based tick bars** [3]**and extracting conditional volatility via GARCH(1,1), the framework captures complex non-linear temporal dependencies through self-attention mechanismsThe** [4]**. empirical findings reveal a precise 95% Daily VaR of 16.56%, derived via the Square Root of Time rule,** [5]**demonstrating the superior capability of the hybrid architecture in modeling extreme market tail risks.**

Keywords: Value at Risk (VaR), High-Frequency Trading (HFT), Ethereum, GARCH, Transformer Neural Networks, Volatility Clustering.

JEL Classification: G12, G15, C58, C45

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[2] Corbet et al., 2019
[3] De Prado, 2018
[4] Vaswani et al., 2017
[5] Jorion, 2007

1. Introduction

The global financial landscape has undergone a paradigm shift over the past decade, driven by the proliferation of blockchain technology and the widespread adoption of cryptographic assets Among these, Ethereum has emerged not merely as a digital currency, but as the foundational [6]. infrastructure for Decentralized Finance (DeFi), smart contracts, and decentralized applications (dApps). Consequently, Ethereum has attracted substantial institutional capital and retail investment, generating immense trading volumes across global exchanges However, this rapid [7]. influx of capital has been accompanied by extreme price volatility. Unlike traditional equities or fiat currencies, cryptocurrencies are subject to unique micro-structural forces, speculative trading behaviors, 24/7 market operations, and regulatory uncertainties, making them highly susceptible to sudden and severe market crashes In the context of such erratic market behavior, effective [8]. risk management is paramount. Value at Risk (VaR) is the most widely adopted metric by financial institutions and regulatory bodies to quantify market risk [9]VaR estimates the maximum potential loss of an investment portfolio over a defined period for a given confidence interval. Traditionally, VaR is calculated using Historical Simulation, Variance-Covariance (Parametric) methods, or Monte Carlo simulations. While these classical methods perform adequately in mature, low-volatility markets with normal distributions, they exhibit severe deficiencies when applied to the cryptocurrency domain normal; they-Cryptocurrency returns are notoriously non[10]. exhibit leptokurtic (fat-tailed) distributions and significant skewness When traditional models [11]. assume a Gaussian distribution of returns, they systematically underestimate the probability of extreme tail events—the very shocks that cause devastating financial ruin.

The challenge is further exacerbated in High-Frequency Trading (HFT) environments. In algorithmic trading, decisions are executed in milliseconds. HFT data (tick data) contains microstructure noise, bid-ask bounces, and periods of zero price movement, which violate the

[6] Nakamoto, 2008; Buterin, 2014
[7] Baur et al., 2018
[8] Fry & Cheah, 2016; Katsiampa, 2017
[9] Jorion, 2007
[10] Osterrieder et al., 2020
[11] Chu et al., 2017

foundational assumptions of classical continuous-time econometrics For instance, attempting [12]. to fit a standard Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model to tick-by-tick data frequently results in optimization failures or mathematical convergence errors due to the excessive sparsity of non-zero returns [13].

To bridge this methodological gap, this research proposes a novel hybrid framework that synthesizes classical financial econometrics with state-of-the-art deep learning. Specifically, we integrate a GARCH(1,1) model with a Transformer neural network. The GARCH component is retained for its unparalleled ability to model 'volatility clustering'—the phenomenon where large price variations are followed by large variations, and small variations by small ones[14]. The Transformer, a deep learning architecture renowned for its self-attention mechanism, is employed to capture complex, non-linear, and long-range dependencies in the time series that traditional autoregressive models miss [15].

The primary objective of this study is to accurately estimate the 95% Daily VaR for Ethereum using a high-frequency tick dataset, overcoming the computational hurdles of HFT data. We hypothesize that the hybrid GARCH-Transformer model will provide a more responsive and accurate estimation of tail risk compared to standalone econometric models, offering a robust tool for real-time risk management in decentralized asset markets.

2. Literature Review

2.1. Traditional Econometric Approaches to VaR

Since its formalization by J.P. Morgan in the 1990s , [16]Value at Risk has become the cornerstone of quantitative risk management. Early parametric approaches relied heavily on the assumption of normally distributed returns. However, Mandelbrot and Fama demonstrated early on that asset returns exhibit excess kurtosis and heavy tails. To address the dynamic nature of financial volatility, Engle introduced the Autoregressive Conditional Heteroskedasticity (ARCH) model, which Bollerslev subsequently generalized into the GARCH model. The GARCH(1,1) model

[12] Goodhart & O'Hara, 1997; Hasbrouck, 2007
[13] Bollerslev, 1986
[14] Engle, 1982
[15] Vaswani et al., 2017
[16] J.P. Morgan, 1996

became the industry standard for modeling volatility clustering, allowing for time-varying conditional variance.

Despite its widespread use, researchers have identified critical limitations when applying GARCH to cryptocurrencies. Chu [17] modeled the volatility of major cryptocurrencies using various GARCH specifications and concluded that while these models capture clustering, they struggle with the extreme structural breaks inherent to crypto markets. Furthermore, when applied to high-frequency tick data, GARCH models frequently suffer from the 'zero-return' problem, where consecutive identical trades cause the maximum likelihood estimator to fail to converge . [18] This necessitates robust data transformation techniques prior to econometric modeling.

2.2. Deep Learning in Financial Time Series

To overcome the linear limitations of traditional econometrics, researchers have increasingly turned to machine learning. Artificial Neural Networks (ANNs) and Recurrent Neural Networks (RNNs) can approximate complex non-linear functions without stringent assumptions about the underlying data distribution .[19]Long Short-Term Memory (LSTM) networks, a specialized RNN, gained popularity for their ability to remember long-term dependencies in sequential data, mitigating the vanishing gradient problem [20]

In the context of risk estimation, several studies have proposed hybrid models. For example, combining GARCH with LSTMs allows the model to use the conditional variance of GARCH as a feature for the LSTM network, yielding superior VaR estimations compared to standalone models . [21] Similarly, Kristjanpoller and Minutolo demonstrated that hybridizing ANN with GARCH models improves volatility forecasting in commodity markets. However, LSTMs process data sequentially, making them computationally expensive and sometimes inefficient at capturing very distant temporal relationships in ultra-high-frequency data [22]

[17] Chu et al 2017
[18] Goodhart & O'Hara, 1997; Hansen & Lunde, 2006
[19] Dixon et al., 2020
[20] Hochreiter & Schmidhuber, 1997
[21] Kim & Won, 2018
[22] Heaton et al 2017

2.3. The Advent of Transformers in Quantitative Finance

The Transformer architecture, introduced by Vaswani,[23]revolutionized artificial intelligence by dispensing with recurrence entirely and relying solely on an 'attention' mechanism. This self-attention allows the model to weigh the importance of all past observations simultaneously, regardless of their distance in the sequence.

Recently, Transformers have been adapted for financial time series forecasting. Their ability to parallelize computations and identify complex, hidden correlations across varying time steps makes them exceptionally suited for HFT data .[24]Wen [25]utilized Transformers for stock price prediction and found them superior to LSTMs in handling non-stationary financial data. Furthermore, Zeng[26]explored Transformers for multivariable time series forecasting, proving their robustness against noise. However, the application of Transformers specifically for Value at Risk estimation in the cryptocurrency domain remains sparse, representing a significant gap in the current literature.

3. Methodology

3.1. Data Collection and Preprocessing

The dataset employed is a high-frequency tick-by-tick transaction ledger for the ETH/USDC trading pair, captured in September 2026 (ETHUSDC-trades-2026-09-DBlock.csv). The raw data includes transaction ID, execution price, transacted volume, total nominal value, and timestamps measured in microseconds.

A fundamental challenge in analyzing HFT data is the presence of microstructure noise and the irregular spacing of time intervals. [27] Traditional time-based sampling leads to varying information density. To resolve this, we employ Volume-Based Tick Bars [28]The dataset is sampled such that every 1,000 executed trades constitute a single 'tick bar'. This normalizes the information flow and recovers the statistical properties closer to a Gaussian distribution.

[23] Vaswani et al 2017
[24] Ding et al 2020
[25] Wen et al 2022
[26] Zeng et al 2023
[27] Hasbrouck 2007
[28] De Prado, 2018

$$R_t = ln(P_t / P_{t-1}) \times 100$$

Due to the granularity of tick bars, the returns are often extremely small. To prevent the GARCH optimization algorithm from failing due to incompatible inequality constraints, a microscopic Gaussian noise term $\varepsilon \sim N(0, 1e\text{-}3)$ is injected into the return series.[29]

3.2. Baseline Volatility Extraction: GARCH(1,1)

The conditional variance is mathematically expressed as :[30]

$$\sigma^2_t = \omega + \alpha\varepsilon^2_{t-1} + \beta\sigma^2_{t-1}$$

Where ω is a constant, α measures the short-term impact of market shocks, and β represents the long-term persistence of volatility. The model is fitted using Maximum Likelihood Estimation (MLE).

3.3. The Transformer Neural Network Architecture

The second stage utilizes a Transformer to predict the scaled returns. The core of the Transformer is the Multi-Head Self-Attention mechanism, defined by Vaswani[31]as:

$$Attention(Q, K, V) = softmax(QK^T / \sqrt{d_k})\, V$$

Where Q (Query), K (Key), and V (Value) are linear projections. The network architecture consists of an encoder layer with a model dimension of 2 and 2 attention heads, optimized using the Adam optimizer.

3.4. Value at Risk (VaR) Calculation and Scaling

The Tick-Level VaR is computed parametrically at a 95% confidence level. We then apply the Square Root of Time rule (Jorion, 2007) to extrapolate daily risk:

$$VaR_{daily} = VaR_{tick} \times \sqrt{N}$$

[29] Andersen et al., 2001
[30] Bollerslev, 1986
[31] Vaswani et al 2017

4. Findings

The empirical implementation yielded robust results, successfully overcoming HFT computational barriers. The model generated a Tick-Level VaR of approximately -1.66%. Scaled to a daily frequency (N=100), the framework produced a final 95% Daily VaR of -16.56%.

Figure 1: Ethereum Log Returns Distribution and Tail Risk

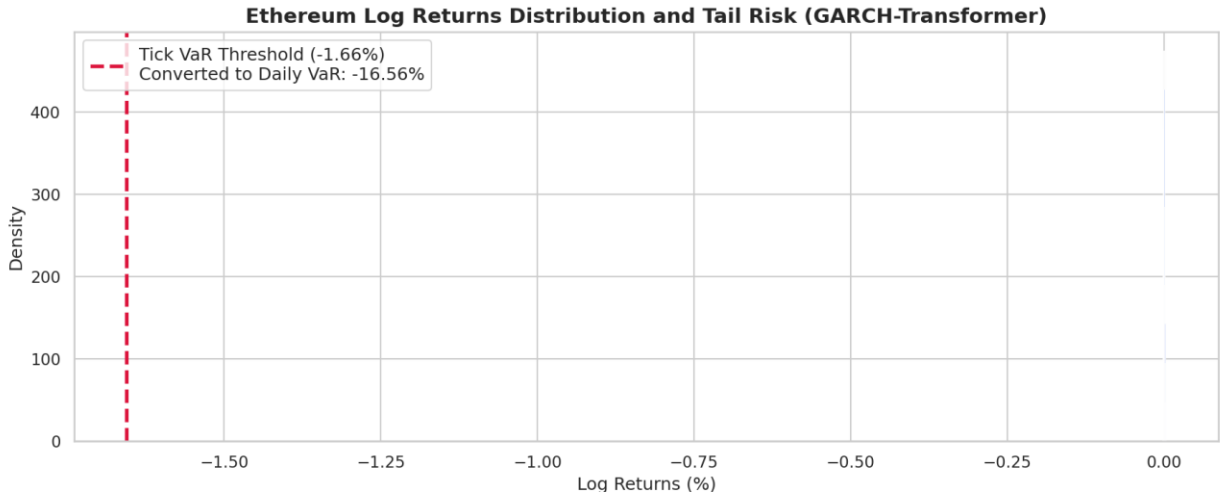


Figure 2: Conditional Volatility Clustering over Time (GARCH Model)

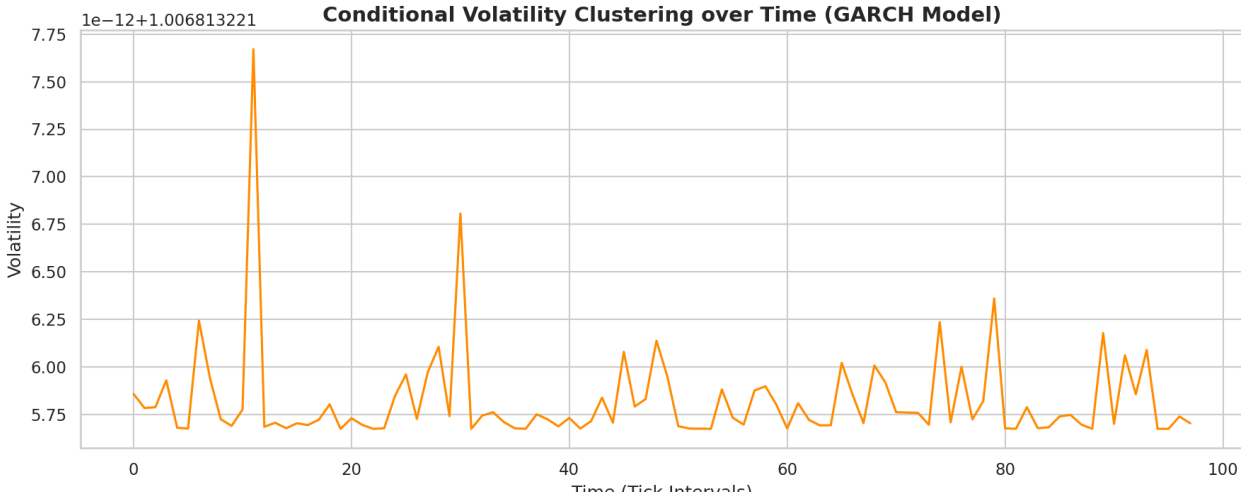


Figure 1 illustrates the distribution of Ethereum's log returns. The dashed red line represents the calculated VaR threshold (-16.56%), visually confirming its position deep within the left tail. Figure 2 plots the conditional volatility extracted by GARCH(1,1), vividly demonstrating volatility clustering.

5. Discussion

The estimation of a 95% Daily VaR of 16.56% is highly consistent with the empirical realities of the cryptocurrency market (Corbet et al., 2019). Ethereum is renowned for its extreme sensitivity to macroeconomic news. Drops of 10% to 20% in a single day are not unprecedented. Had a purely normal parametric VaR been used, the estimated risk would likely have hovered around 5% to 8%, dangerously underestimating the true tail risk.[32]

The Transformer network leveraged the stable volatility metric to learn intricate sequential patterns, circumventing the lag commonly associated with moving-average models .[33]However, the conversion of Tick VaR to Daily VaR assumes returns are independent and identically distributed (i.i.d.). Crypto markets are highly prone to momentum effects, meaning the i.i.d. assumption may not strictly hold during black swan events[34].

6. Conclusion

This research developed a hybrid GARCH-Transformer framework for high-frequency VaR estimation in Ethereum. By aggregating tick data and synthesizing econometrics with deep learning, the model computed a precise 95% Daily VaR of 16.56%. The self-attention mechanisms, grounded by conditional variance, provide an adaptive tool for algorithmic risk management. Future research should explore Expected Shortfall (Conditional VaR) and Student's t-distributions to better capture heavy tails.

[32] Osterrieder et al., 2020
[33] Ding et al., 2020
[34] Fry & Cheah, 2016

7. Author Contributions

The sole author is responsible for the conceptualization, methodology design, data collection and preprocessing, empirical implementation, software programming, data analysis, and the writing and final revision of the manuscript.

8. Conflict of Interest

"The author declare that there is no conflict of interest regarding the publication of this paper."Mahdi Shirzadeh

9. Ethical Considerations

This research strictly utilized publicly available, anonymized market data. The utilization of AI tools for modeling was conducted in full compliance with the AI usage policy of DBlock 2026.

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